

Lab Experiment 8: Conduct Z test for the Data Given using Python Objective:

To test whether the average weight of a species of birds differs from 150 grams.

Procedure:

1. **Null Hypothesis (H_0):** The average weight of the birds is 150 grams.
2. **Alternative Hypothesis (H_1):** The average weight of the birds is not 150 grams.
3. **Sample:** Measure the weights of 30 birds randomly selected from the population.
4. **Z-Test:** Conduct a Z-test to compare the sample mean to 150 grams.
5. **Decision Rule:** Use a significance level of $\alpha = 0.05$.

Python Code Example:

python

Copy code

```
import numpy as np
```

```
import scipy.stats as stats
```

```
# Define the sample data (hypothetical weights in grams)
```

```
sample_data = np.array([152, 148, 151, 149, 147, 153, 150, 148, 152, 149, 151, 150, 149, 152, 151, 148, 150, 152, 149, 150, 148, 153, 151, 150, 149, 152, 148, 151, 150, 153])
```

```
# Population mean under the null hypothesis
```

```
population_mean = 150
```

```
# Calculate sample statistics
```

```
sample_mean = np.mean(sample_data)
```

```
sample_std = np.std(sample_data, ddof=1) # Using sample standard deviation
```

```
# Number of observations
```

```
n = len(sample_data)
```

```
# Calculate the Z-statistic
```

```
z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))
```

```
# Calculate the p-value
```

```
p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic))) # Two-tailed test

# Print results

print(f"Sample Mean: {sample_mean:.2f}")

print(f"Z-Statistic: {z_statistic:.4f}")

print(f"P-Value: {p_value:.4f}")

# Decision based on the significance level

alpha = 0.05

if p_value < alpha:

    print("Reject the null hypothesis: The average weight is significantly different from 150 grams.")

else:

    print("Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.")
```

OUTPUT:

Sample Mean: 150.20

Z-Statistic: 0.6406

P-Value: 0.5218

Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.